HODGE LOCI, MUMFORD–TATE GROUPS, AND THE HODGE CONJECTURE FOR ABELIAN VARIETIES

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These are notes for an introductory talk on Hodge (or Noether–Lefschetz) loci, Mumford–Tate groups, and the Hodge conjecture, with abelian varieties as the running example. The guiding principle is

Hodge classes on A = invariants of the Mumford–Tate group of A,

so that computing Mumford–Tate groups computes Hodge classes. Hodge loci are where Hodge classes "jump", and on the moduli space of abelian varieties they correspond to algebraic subgroups. We end by applying this to the fibers of a maximally varying Lagrangian fibration (Theorems 6.4 and 6.6).

Throughout, varieties are over $\mathbb{C}$, and cohomology means singular cohomology of the associated complex analytic space.

1. Algebraic cycles and the Hodge conjecture

Let X be a smooth projective variety of dimension n.

Definition 1.1. An *algebraic cycle of codimension p* on X is a finite formal sum $Z = \sum_i n_i Z_i$ with $n_i \in \mathbb{Z}$ and $Z_i \subseteq X$ irreducible closed subvarieties of codimension p. These form a free abelian group $\mathcal{Z}^p(X)$.

Definition 1.2. A cycle $Z \in \mathcal{Z}^p(X)$ is *rationally equivalent to zero* if there are subvarieties $W_j \subseteq X$ of codimension $p-1$ and rational functions $f_j \in \mathbb{C}(W_j)^\times$ with $Z = \sum_j \operatorname{div}(f_j)$. The *Chow group* is $\mathrm{CH}^p(X) = \mathcal{Z}^p(X)/\sim_{\mathrm{rat}}$.

For example $\mathrm{CH}^1(X) = \mathrm{Pic}(X)$. Equivalently, $Z \sim_{\mathrm{rat}} 0$ if Z is a sum of differences $\Gamma_0 - \Gamma_\infty$ of fibers of subvarieties $\Gamma \subseteq X \times \mathbb{P}^1$ flat over $\mathbb{P}^1$, so rationally equivalent cycles are deformations of each other over $\mathbb{P}^1$.

1.1. **The cycle class map.** If $Z \subseteq X$ is irreducible of codimension p, integration over the smooth locus of Z gives a well-defined functional $\eta \mapsto \int_Z \eta$ on $H^{2n-2p}(X, \mathbb{C})$ (Lelong), and hence by Poincaré duality a class $[Z] \in H^{2p}(X, \mathbb{C})$. In fact $[Z] \in H^{2p}(X, \mathbb{Z})$: it is the Poincaré dual of the fundamental class of Z. Since cohomology classes are locally constant in families, this descends to the *cycle class map*

$$\mathrm{cl}\colon \mathrm{CH}^p(X) \to H^{2p}(X, \mathbb{Z}).$$

Lemma 1.3. *For any cycle Z, the image of $[Z]$ in $H^{2p}(X, \mathbb{C})$ lies in $H^{p,p}(X)$.*

Proof. By Poincaré duality it suffices to show $\int_Z \eta = 0$ for every closed form η of type (a, b) with $a + b = 2n - 2p$ and $(a, b) \neq (n-p, n-p)$. But then $a > n - p$ or $b > n - p$, and an (a, b)-form restricts to zero on the complex manifold Z^{sm}, which has dimension $n - p$. $\square$

Definition 1.4. The groups of *integral* and *rational Hodge classes* are

$$\mathrm{Hdg}^p(X,\mathbb{Z}) = \{\alpha \in H^{2p}(X,\mathbb{Z}) : \alpha_{\mathbb{C}} \in H^{p,p}(X)\},$$
$$\mathrm{Hdg}^p(X) = \mathrm{Hdg}^p(X,\mathbb{Q}) = H^{2p}(X,\mathbb{Q}) \cap H^{p,p}(X).$$

A Hodge class is *algebraic* if it lies in the image of cl (respectively of $\mathrm{cl} \otimes \mathbb{Q}$).

Remark 1.5. The Hodge conjecture only concerns the *image* of cl, so rational equivalence plays no real role below. It is just the most natural equivalence relation to put on cycles.

1.2. The (integral) Hodge conjecture.

Conjecture 1.6 (Integral Hodge conjecture)**.** *For all p, the cycle class map*

$$\mathrm{cl}\colon \mathrm{CH}^p(X) \to \mathrm{Hdg}^p(X,\mathbb{Z})$$

is surjective.

Theorem 1.7 (Lefschetz $(1,1)$)**.** *The integral Hodge conjecture holds for $p = 1$: every class in $\mathrm{Hdg}^1(X,\mathbb{Z})$ is the first Chern class of a line bundle.*

It also holds trivially for $p = 0$ and $p = n$. Unfortunately, it is false in general.

Theorem 1.8 (Atiyah–Hirzebruch [AH62])**.** *There are smooth projective X with torsion classes in $H^4(X,\mathbb{Z})$ that are not algebraic. (Torsion classes vanish in $H^4(X,\mathbb{C})$, so they are automatically Hodge classes.)*

Theorem 1.9 (Kollár [Kol92])**.** *Let $X \subseteq \mathbb{P}^4$ be a very general hypersurface of degree d, where $p^3 \mid d$ for some prime $p \geq 5$. Then every curve on X has degree divisible by p.*

By the Lefschetz hyperplane theorem and Poincaré duality, $H^4(X,\mathbb{Z}) \cong \mathbb{Z}$ is generated by a class ℓ with $\ell \cdot h = 1$, and every class in $H^4(X,\mathbb{Z})$ is Hodge. So ℓ is a non-torsion Hodge class that is not algebraic, even though $d\ell = h^2$ is. The standard fix is to pass to $\mathbb{Q}$-coefficients.

Conjecture 1.10 (Hodge conjecture)**.** *For all p, $\mathrm{Hdg}^p(X)$ is spanned over $\mathbb{Q}$ by classes of algebraic cycles, i.e. $\mathrm{cl} \otimes \mathbb{Q}\colon \mathrm{CH}^p(X)_{\mathbb{Q}} \to \mathrm{Hdg}^p(X)$ is surjective.*

Proposition 1.11. *The Hodge conjecture holds in degrees $p = 0, 1, n-1, n$. In particular it holds for all X of dimension at most 3.*

Proof. The cases $p = 0, n$ are trivial and $p = 1$ is Lefschetz $(1,1)$. For $p = n-1$, let L be cup product with the class of a hyperplane section H. By hard Lefschetz, $L^{n-2}\colon H^2(X,\mathbb{Q}) \to H^{2n-2}(X,\mathbb{Q})$ is an isomorphism, and it maps $H^{1,1}$ onto $H^{n-1,n-1}$, so it gives an isomorphism $\mathrm{Hdg}^1(X) \cong \mathrm{Hdg}^{n-1}(X)$. It sends $[D]$ to $[D \cap H_1 \cap \cdots \cap H_{n-2}]$ for general hyperplane sections H_i, so it preserves algebraic classes. □

Remark 1.12. If A is an abelian variety, then $H^*(A,\mathbb{Z})$ is an exterior algebra on $H^1(A,\mathbb{Z})$, and in particular torsion-free. So there are no Atiyah–Hirzebruch type counterexamples on abelian varieties. Nevertheless, the integral Hodge conjecture fails for very general principally polarized abelian varieties of dimension ≥ 4 by recent work of Engel, de Gaay Fortman, and Schreieder; see Theorem 5.5 below.

2. Abelian varieties and their moduli

2.1. **Hodge structures.** Let $\mathbb{S} = \operatorname{Res}_{\mathbb{C}/\mathbb{R}} \mathbb{G}_m$ be the Deligne torus, so $\mathbb{S}(\mathbb{R}) = \mathbb{C}^\times$. A $\mathbb{Q}$-*Hodge structure of weight* w is a finite-dimensional $\mathbb{Q}$-vector space V together with a homomorphism of real algebraic groups $h\colon \mathbb{S} \to \mathrm{GL}(V_\mathbb{R})$ such that

$$V_\mathbb{C} = \bigoplus_{p+q=w} V^{p,q}, \qquad h(z) \text{ acts on } V^{p,q} \text{ by } z^p \overline{z}^q.$$

(Conventions differ by a sign, which will not matter.) Two consequences we will use constantly:

- for $r \in \mathbb{R}^\times$, $h(r) = r^w$ is a scalar;
- for $z \in U(1) = \{|z| = 1\}$, $h(z)$ acts on $V^{p,q}$ by z^{p-q}.

Duals, tensor products, and exterior powers of Hodge structures are Hodge structures; we write $T^{a,b}(V) = V^{\otimes a} \otimes (V^\vee)^{\otimes b}$, of weight $(a-b)w$, and call $T^{a,b}(V)$, $\Lambda^k V$, $\operatorname{Sym}^k V$ and sums of these *tensor constructions* of V. If W has weight $2p$, a *Hodge class* in W is an element of $W \cap W^{p,p}$.

2.2. **Abelian varieties.** A complex abelian variety of dimension g is a projective complex torus $A = \mathbb{C}^g/\Lambda$. We always write

$$V = H^1(A, \mathbb{Q}).$$

Then $H^k(A,\mathbb{Z}) = \Lambda^k H^1(A,\mathbb{Z})$, $H^1(A,\mathbb{C}) = H^{1,0} \oplus H^{0,1}$ with $H^{1,0} = H^0(A, \Omega^1_A)$, and $H^{p,q}(A) = \Lambda^p H^{1,0} \otimes \Lambda^q H^{0,1}$. So:

The Hodge structure on $H^*(A,\mathbb{Q}) = \Lambda^\bullet V$ is determined by the weight one Hodge structure V,

and $\mathrm{Hdg}^p(A)$ is the set of Hodge classes in $\Lambda^{2p} V$.

A polarization is the class $\theta = c_1(L) \in H^2(A,\mathbb{Z})$ of an ample line bundle. We will only consider principal polarizations, for which there is a basis $e_1, \dots, e_g, f_1, \dots, f_g$ of $H^1(A,\mathbb{Z})$ with

$$\theta = \sum_{i=1}^g e_i \wedge f_i \in \Lambda^2 H^1(A,\mathbb{Z}).$$

Let ψ be the symplectic form on V for which (e_i, f_i) is a symplectic basis. We write

$$\mathrm{Sp}(V) = \mathrm{Sp}(V,\psi) = \mathrm{Stab}_{\mathrm{GL}(V)}(\theta) \cong \mathrm{Sp}_{2g,\mathbb{Q}}, \qquad \mathrm{GSp}(V) = \mathbb{G}_m \cdot \mathrm{Sp}(V),$$

the latter being the group of $g \in \mathrm{GL}(V)$ that preserve ψ up to a scalar.

2.3. **The Siegel upper half space and $\mathcal{A}_g$.** Fix $V_\mathbb{Z} = \mathbb{Z}^{2g}$ with its standard symplectic form ψ, and let $V = V_\mathbb{Z} \otimes \mathbb{Q}$. Let

$$\mathfrak{H}_g = \left\{ h\colon \mathbb{S} \to \mathrm{GL}(V_\mathbb{R}) \;\middle|\; \begin{array}{l} h \text{ is a Hodge structure of weight 1 with} \\ \psi(h(i)x, h(i)y) = \psi(x,y), \text{ and } \psi(x, h(i)y) \text{ positive definite} \end{array} \right\}.$$

The last two conditions say that ψ is a polarization of the Hodge structure (for a suitable sign convention).

Theorem 2.1 (Riemann)**.** *Points of $\mathfrak{H}_g$ are in bijection with isomorphism classes of pairs (A, β), where A is a principally polarized abelian variety of dimension g and $\beta\colon H^1(A,\mathbb{Z}) \cong V_\mathbb{Z}$ is a symplectic isomorphism.*

The group $\mathrm{Sp}(V_{\mathbb{R}}) = \mathrm{Sp}_{2g}(\mathbb{R})$ acts on $\mathfrak{H}_g$ by conjugation, $g \cdot h = ghg^{-1}$, and this action is transitive with stabilizer isomorphic to $U(g)$. So

$$\mathfrak{H}_g \cong \mathrm{Sp}_{2g}(\mathbb{R})/U(g) \cong \{\tau \in M_g(\mathbb{C}) : \tau = \tau^T,\ \mathrm{Im}(\tau) > 0\},$$

the classical Siegel upper half space, of complex dimension $g(g+1)/2$. Forgetting the marking β, we get the moduli space of principally polarized abelian varieties

$$\mathcal{A}_g = \mathrm{Sp}_{2g}(\mathbb{Z})\backslash\mathfrak{H}_g,$$

which is a quasi-projective variety of dimension $g(g+1)/2$ (Baily–Borel).

Remark 2.2 (Shimura varieties). $\mathcal{A}_g$ is the basic example of a *Shimura variety*: a quotient $\Gamma\backslash X$, where G is a reductive group over $\mathbb{Q}$, X is a $G(\mathbb{R})$-conjugacy class of homomorphisms $\mathbb{S} \to G_{\mathbb{R}}$ (satisfying axioms that make X a Hermitian symmetric domain), and $\Gamma \subseteq G(\mathbb{Q})$ is an arithmetic subgroup. Here $G = \mathrm{GSp}(V)$ and $X = \mathfrak{H}_g$. The point is that the period domain is a single orbit of a real algebraic group acting by conjugation on Hodge structures. We will see that the Hodge loci in $\mathcal{A}_g$ have the same form, for smaller groups (Section 4).

The key feature of $\mathfrak{H}_g$ is that for every $h \in \mathfrak{H}_g$,

$$H^{2p}(A_h, \mathbb{Q}) = \Lambda^{2p} V$$

is the *same* $\mathbb{Q}$-vector space; only the Hodge decomposition, induced by h, moves. This makes Hodge loci in $\mathfrak{H}_g$ very concrete.

3. Mumford–Tate groups

Definition 3.1. Let V be a $\mathbb{Q}$-Hodge structure given by $h\colon \mathbb{S} \to \mathrm{GL}(V_{\mathbb{R}})$.

- The *Mumford–Tate group* $\mathrm{MT}(V)$ is the smallest $\mathbb{Q}$-algebraic subgroup $G \subseteq \mathrm{GL}(V)$ with $h(\mathbb{C}^\times) \subseteq G(\mathbb{R})$.
- The *Hodge group* (or special Mumford–Tate group) $\mathrm{Hg}(V)$ is the smallest $\mathbb{Q}$-algebraic subgroup $G \subseteq \mathrm{GL}(V)$ with $h(U(1)) \subseteq G(\mathbb{R})$.

For an abelian variety we write $\mathrm{MT}(A) = \mathrm{MT}(H^1(A, \mathbb{Q}))$ and $\mathrm{Hg}(A) = \mathrm{Hg}(H^1(A, \mathbb{Q}))$.

Remark 3.2.
(1) $\mathrm{MT}(V)$ and $\mathrm{Hg}(V)$ are connected, since the identity component also satisfies the defining property.
(2) If V has weight $w \neq 0$, then $\mathrm{MT}(V) = \mathbb{G}_m \cdot \mathrm{Hg}(V)$, since $\mathbb{C}^\times = \mathbb{R}_{>0} \cdot U(1)$ and $h(r) = r^w$ is a scalar.
(3) If V is polarizable (e.g. $V = H^1(A, \mathbb{Q})$), then $\mathrm{MT}(V)$ and $\mathrm{Hg}(V)$ are reductive [Del82, Moo04].
(4) For an abelian variety, $\mathrm{Hg}(A^m) = \mathrm{Hg}(A)$, acting diagonally on $H^1(A^m, \mathbb{Q}) = V^{\oplus m}$: the diagonal copy of $\mathrm{Hg}(A)$ contains $h(U(1))$, and any $\mathbb{Q}$-subgroup of it containing $h(U(1))$ projects onto $\mathrm{Hg}(A)$.

The reason to care about Hg is the following.

Proposition 3.3. *Let W be a tensor construction of V of even weight $2p$. Then $t \in W$ is a Hodge class if and only if t is fixed by $\mathrm{Hg}(V)$. In particular, for an abelian variety A,*

$$\mathrm{Hdg}^p(A) = (\Lambda^{2p} V)^{\mathrm{Hg}(A)}.$$

Proof. Since $\mathrm{Hg}(V) \subseteq \mathrm{GL}(V)$, it acts on every tensor construction of V. An element $z \in U(1)$ acts on $W^{a,b}$ by z^{a-b}, so $t \in W^{p,p}$ if and only if $h(z)t = t$ for all $z \in U(1)$. Since t is rational, its stabilizer $\mathrm{Stab}(t) \subseteq \mathrm{GL}(V)$ is a $\mathbb{Q}$-algebraic subgroup, and so

$$t \in W^{p,p} \iff h(U(1)) \subseteq \mathrm{Stab}(t)(\mathbb{R}) \iff \mathrm{Hg}(V) \subseteq \mathrm{Stab}(t)$$

by minimality of $\mathrm{Hg}(V)$. □

Remark 3.4. The Mumford–Tate group itself acts on a Hodge class of weight $2p$ through the character $\mathbb{G}_m \to \mathbb{G}_m$, $r \mapsto r^{2p}$ on the central $\mathbb{G}_m$, which is why it is more convenient to work with Hg. (Alternatively, Tate twist everything to weight 0.)

Corollary 3.5. *Let A be a principally polarized abelian variety, $V = H^1(A, \mathbb{Q})$.*

(a) Since $\theta \in \Lambda^2 V$ is a Hodge class, $\mathrm{Hg}(A) \subseteq \mathrm{Sp}(V)$ *and* $\mathrm{MT}(A) \subseteq \mathrm{GSp}(V)$.
(b) Sub-Hodge structures of V are exactly the $\mathrm{Hg}(A)$*-stable subspaces.*
(c) Morphisms of Hodge structures are Hodge classes in $\mathrm{Hom}(V, V') = V^\vee \otimes V'$ (weight 0). So $\mathrm{End}_{\mathrm{HS}}(V) = \mathrm{End}(V)^{\mathrm{Hg}(A)}$, and more generally, since $\mathrm{Hom}(A, B) \otimes \mathbb{Q} \cong \mathrm{Hom}_{\mathrm{HS}}(H^1(B, \mathbb{Q}), H^1(A, \mathbb{Q}))$, endomorphisms and isogenies of abelian varieties are detected by Hg.

In fact $\mathrm{Hg}(V)$ is *determined* by the Hodge classes in all tensor constructions:

Fact 3.6 (Deligne [Del82])**.** *If V is polarizable, then* $\mathrm{Hg}(V)$ *is the subgroup of* $\mathrm{GL}(V)$ *fixing every Hodge class in every $T^{a,b}(V)$.*

3.1. **Elliptic curves.** For an elliptic curve E, $V = H^1(E, \mathbb{Q})$ is 2-dimensional, so $\mathrm{Sp}(V) = \mathrm{SL}(V)$.

Proposition 3.7. *Let E be an elliptic curve and $V = H^1(E, \mathbb{Q})$.*

(i) If E does not have complex multiplication, then $\mathrm{Hg}(E) = \mathrm{SL}(V) \cong \mathrm{SL}_2$ *and* $\mathrm{MT}(E) = \mathrm{GL}(V) \cong \mathrm{GL}_2$.
(ii) If $\mathrm{End}(E) \otimes \mathbb{Q} = K$ is an imaginary quadratic field, then V is a 1-dimensional K-vector space,

$$\mathrm{MT}(E) = T_K := \mathrm{Res}_{K/\mathbb{Q}} \mathbb{G}_m, \qquad \mathrm{Hg}(E) = U_K := \ker\left(\mathrm{Nm}\colon T_K \to \mathbb{G}_m\right).$$

Proof. $\mathrm{Hg} = \mathrm{Hg}(E)$ is a connected reductive subgroup of $\mathrm{SL}(V)$, and it is nontrivial since $h(z)$ acts on $V^{1,0}$ by z. The connected reductive subgroups of SL_2 are 1, the 1-dimensional tori, and SL_2 itself.

Suppose $\mathrm{Hg} = T$ is a torus. Over $\mathbb{C}$, T acts on $V_\mathbb{C}$ through two distinct characters χ, χ^{-1} (the only scalars in SL_2 are ± 1), so its centralizer $K' = \mathrm{End}(V)^T$ is a 2-dimensional commutative $\mathbb{Q}$-algebra. By Theorem 3.5(c), $K' \subseteq \mathrm{End}(E) \otimes \mathbb{Q}$. Since $\mathrm{End}(E) \otimes \mathbb{Q}$ is either $\mathbb{Q}$ or an imaginary quadratic field, E has CM.

Conversely, suppose E has CM by K. Then K acts on V commuting with Hg, so Hg lies in the centralizer of K in $\mathrm{SL}(V)$, which is $T_K \cap \mathrm{SL}(V) = U_K$. This is a 1-dimensional torus, and Hg is nontrivial and connected, so $\mathrm{Hg} = U_K$. Finally $\mathrm{MT}(E) = \mathbb{G}_m \cdot U_K = T_K$. □

Remark 3.8. $E_\tau = \mathbb{C}/(\mathbb{Z} + \tau\mathbb{Z})$ has CM if and only if $\tau \in \mathfrak{H}_1$ is an imaginary quadratic number. So CM elliptic curves correspond to a *countable dense* subset of $\mathfrak{H}_1$.

Example 3.9 (Hodge classes on $E \times E$)**.** We have $H^1(E \times E) = V \oplus V$, and by Theorem 3.2(4), $\mathrm{Hg}(E \times E) = \mathrm{Hg}(E)$ acting diagonally. Then

$$H^2(E \times E, \mathbb{Q}) = \Lambda^2 V \oplus (V \otimes V) \oplus \Lambda^2 V.$$

$\mathrm{Hg}(E) \subseteq \mathrm{SL}(V)$ acts trivially on each $\Lambda^2 V$, giving the classes of $E \times 0$ and $0 \times E$. Using the $\mathrm{GL}(V)$-equivariant isomorphism $V \cong V^\vee \otimes \Lambda^2 V$, we get $V \otimes V \cong \mathrm{End}(V)$ as $\mathrm{Hg}(E)$-representations, so $(V \otimes V)^{\mathrm{Hg}(E)} \cong \mathrm{End}(V)^{\mathrm{Hg}(E)} = \mathrm{End}(E) \otimes \mathbb{Q}$. Therefore

$$\rho(E \times E) = 2 + \dim_{\mathbb{Q}} \mathrm{End}(E) \otimes \mathbb{Q} = \begin{cases} 3 & E \text{ without CM}, \\ 4 & E \text{ with CM}. \end{cases}$$

The extra classes are the diagonal and, in the CM case, the graph of a complex multiplication. So "E has CM" is detected by a Hodge class on $E \times E$.

Proposition 3.10. *Let $E_1, \dots, E_n$ be pairwise non-isogenous elliptic curves without CM, $A = E_1 \times \cdots \times E_n$, and $V_i = H^1(E_i, \mathbb{Q})$. Then*

$$\mathrm{Hg}(A) = \mathrm{SL}(V_1) \times \cdots \times \mathrm{SL}(V_n),$$
$$\mathrm{MT}(A) = \mathbb{G}_m \cdot \mathrm{SL}_2^n = \Big\{ (g_i) \in \prod_i \mathrm{GL}(V_i) : \det g_1 = \cdots = \det g_n \Big\}.$$

Proof. $\prod_i \mathrm{Hg}(E_i) = \prod_i \mathrm{SL}(V_i)$ is a $\mathbb{Q}$-group containing $h(U(1)) = \{(h_1(z), \dots, h_n(z))\}$, so $\mathrm{Hg}(A)$ is contained in it. The projection of $\mathrm{Hg}(A)$ to $\mathrm{SL}(V_i)$ is a $\mathbb{Q}$-group containing $h_i(U(1))$, so it is all of $\mathrm{SL}(V_i)$ by Theorem 3.7. Let $\mathfrak{h} \subseteq \bigoplus_i \mathfrak{sl}(V_i)$ be the Lie algebra of $\mathrm{Hg}(A)$.

Take $n = 2$. Then $\mathfrak{h} \cap (\mathfrak{sl}(V_1) \oplus 0)$ is an ideal of $\mathfrak{sl}(V_1)$, because $\mathfrak{h}$ surjects onto $\mathfrak{sl}(V_1)$. As $\mathfrak{sl}_2$ is simple, either it is all of $\mathfrak{sl}(V_1)$, in which case $\mathfrak{h} = \mathfrak{sl}(V_1) \oplus \mathfrak{sl}(V_2)$, or it is 0. Arguing the same way with the second factor, if $\mathfrak{h} \neq \mathfrak{sl}(V_1) \oplus \mathfrak{sl}(V_2)$ then $\mathfrak{h}$ is the graph of an isomorphism $\mathfrak{sl}(V_1) \to \mathfrak{sl}(V_2)$ defined over $\mathbb{Q}$. Every such isomorphism has the form $x \mapsto fxf^{-1}$ for a linear isomorphism $f \colon V_1 \to V_2$, since $\mathrm{Aut}(\mathfrak{sl}_2) = \mathrm{PGL}_2$ and $\mathrm{PGL}_2(\mathbb{Q}) = \mathrm{GL}_2(\mathbb{Q})/\mathbb{Q}^\times$. Then $\mathrm{Hg}(A) = \{(g, fgf^{-1})\}$ fixes $f \in \mathrm{Hom}(V_1, V_2)$, so f is a morphism of Hodge structures by Theorem 3.5(c). This gives an isogeny $E_2 \to E_1$, a contradiction.

For general n, induct: by induction $\mathfrak{h}$ surjects onto $\mathfrak{sl}(V_2) \oplus \cdots \oplus \mathfrak{sl}(V_n)$, and if $\mathfrak{h} \cap (\mathfrak{sl}(V_1) \oplus 0) = 0$, then $\mathfrak{h}$ is the graph of a surjection $\bigoplus_{i \geq 2} \mathfrak{sl}(V_i) \to \mathfrak{sl}(V_1)$. Its kernel is an ideal, hence a sum of all but one factor $\mathfrak{sl}(V_j)$, so the projection of $\mathfrak{h}$ to $\mathfrak{sl}(V_1) \oplus \mathfrak{sl}(V_j)$ is a graph, contradicting the case $n = 2$. □

Note that $\mathrm{SL}_2 = \mathrm{Sp}_2$, so $\mathrm{MT}(E_1 \times \cdots \times E_n) = \mathbb{G}_m \cdot (\mathrm{Sp}_{2,\mathbb{Q}})^n$. This is the shape that will appear in Theorem 6.4.

4. Hodge loci

4.1. **Definitions.** Let $f \colon X \to S$ be a smooth projective morphism to a connected complex manifold, and fix p. Let $\pi \colon \widetilde{S} \to S$ be the universal cover. Over $\widetilde{S}$ the local system $R^{2p} f_* \mathbb{Q}$ is trivial, so parallel transport identifies $H^{2p}(X_{\tilde{s}}, \mathbb{Q})$ with a fixed $\mathbb{Q}$-vector space H for all $\tilde{s} \in \widetilde{S}$. The Hodge filtration $F^\bullet_{\tilde{s}}$ on $H_{\mathbb{C}}$ varies holomorphically with $\tilde{s}$ (Griffiths).

Definition 4.1. For $\lambda \in H$, the *Hodge locus* (or *Noether–Lefschetz locus*) of λ is

$$\mathrm{NL}(\lambda) = \{\tilde{s} \in \widetilde{S} : \lambda \in \mathrm{Hdg}^p(X_{\tilde{s}})\}.$$

Its image $\pi(\mathrm{NL}(\lambda)) \subseteq S$ is the set of s such that some parallel transport of λ to X_s is a Hodge class. The *Hodge locus of f* is

$$\mathrm{HL}(f) = \bigcup_{\lambda \in H,\ \mathrm{NL}(\lambda) \neq \widetilde{S}} \pi(\mathrm{NL}(\lambda)) \subseteq S.$$

The same definitions make sense for any polarizable variation of Hodge structure, e.g. for tensor constructions of $R^k f_* \mathbb{Q}$.

Example 4.2 (Noether–Lefschetz)**.** Let S parametrize smooth surfaces $X_s \subseteq \mathbb{P}^3$ of degree $d \geq 4$, and $p = 1$. By Lefschetz $(1,1)$, $\mathrm{HL}(f)$ is the locus of surfaces with Picard number $\rho(X_s) \geq 2$. The Noether–Lefschetz theorem says that $\mathrm{HL}(f) \neq S$, i.e. the very general such surface has $\mathrm{Pic}(X_s) = \mathbb{Z} \cdot \mathcal{O}(1)$. However, $\mathrm{HL}(f)$ is dense in S in the analytic topology [CHM88]. This is where the name comes from; for classes of higher degree, "Hodge locus" is more common.

Proposition 4.3. $\mathrm{NL}(\lambda)$ *is a closed complex analytic subset of* $\widetilde{S}$.

Proof. Since λ is real and $F^p \cap \overline{F^p} = H^{p,p}$ in weight $2p$, λ is a Hodge class at $\tilde{s}$ if and only if $\lambda \in F^p_{\tilde{s}}$. So $\mathrm{NL}(\lambda)$ is the zero locus of the image of the constant (hence holomorphic) section λ in the holomorphic vector bundle $(H \otimes \mathcal{O}_{\widetilde{S}})/F^p$. □

Much more is true.

Theorem 4.4 (Cattani–Deligne–Kaplan [CDK95])**.** *If S is quasi-projective and f is algebraic, then $\pi(\mathrm{NL}(\lambda))$ is a closed algebraic subvariety of S for every λ.*

Remark 4.5 (Hodge conjecture ⇒ CDK). Suppose the Hodge conjecture holds. Then λ_s is Hodge if and only if it is a $\mathbb{Q}$-linear combination of classes of subvarieties of X_s. The relative Hilbert scheme of X/S has countably many components, each projective over S, and the cohomology class of a subvariety is locally constant in flat families. So the locus where λ is algebraic is a countable union of images of (products of) such components, hence a countable union of closed algebraic subvarieties of S. Thus CDK is exactly what the Hodge conjecture predicts, and is one of the best pieces of evidence for it.

Definition 4.6. A point $s \in S$ is *very general* if $s \notin \mathrm{HL}(f)$.

If $\mathrm{NL}(\lambda) \neq \widetilde{S}$, then $\mathrm{NL}(\lambda)$ is nowhere dense in $\widetilde{S}$, so $\pi(\mathrm{NL}(\lambda))$ is a countable union of nowhere dense subsets of S, and by CDK (when S is quasi-projective) it is a proper closed subvariety. There are countably many $\lambda \in H$, so the very general points are the complement of a countable union of proper closed subvarieties; by the Baire category theorem they are dense. By construction:

Lemma 4.7. *If $s \in S$ is very general, then a class in $H^{2p}(X_s, \mathbb{Q})$ is Hodge if and only if it remains Hodge under parallel transport to every fiber. In words: on a very general fiber, the only Hodge classes are the ones forced by the family.*

4.2. **Hodge loci in $\mathcal{A}_g$.** Over $\mathfrak{H}_g$ everything is explicit, since $H^{2p}(A_h, \mathbb{Q}) = \Lambda^{2p} V$ for every h. For a tensor $t \in T^{a,b}(V)$ (e.g. $t \in \Lambda^{2p} V$), put

$$\mathrm{NL}(t) = \{h \in \mathfrak{H}_g : t \text{ is a Hodge class for } h\}.$$

For a $\mathbb{Q}$-algebraic subgroup $H \subseteq \mathrm{Sp}(V)$, put

$$\mathfrak{H}_g(H) = \{h \in \mathfrak{H}_g : h(U(1)) \subseteq H(\mathbb{R})\} = \{h \in \mathfrak{H}_g : \mathrm{Hg}(A_h) \subseteq H\}.$$

Proposition 4.8. *(i)* $\mathrm{NL}(t) = \mathfrak{H}_g(G_t)$, *where* $G_t = \mathrm{Stab}_{\mathrm{Sp}(V)}(t)$.

(ii) $\mathfrak{H}_g(H)$ *is stable under conjugation by* $H(\mathbb{R})$. *In particular, it contains the orbit* $H(\mathbb{R}) \cdot h$ *of any of its points.*

Proof. (i) is Theorem 3.3, together with the fact that $h(U(1)) \subseteq \mathrm{Sp}(V_{\mathbb{R}})$ for all $h \in \mathfrak{H}_g$. For (ii), if $h(U(1)) \subseteq H(\mathbb{R})$ and $g \in H(\mathbb{R})$, then $gh(U(1))g^{-1} \subseteq H(\mathbb{R})$. □

Fact 4.9 (see [Moo04, Mil05])**.** *Let $h \in \mathfrak{H}_g$ and $H = \mathrm{Hg}(A_h)$. Then $H(\mathbb{R})^+ \cdot h$ is a closed complex submanifold of $\mathfrak{H}_g$ (a Hermitian symmetric domain), and its image in $\mathcal{A}_g$ is a closed algebraic subvariety, called a* special *(or* Shimura*) subvariety. Its very general point has Hodge group exactly H. The irreducible components of Hodge loci in $\mathcal{A}_g$ are exactly the special subvarieties, and there are countably many of them since there are countably many $\mathbb{Q}$-subgroups of $\mathrm{Sp}(V)$.*

So we have a dictionary:

Hodge theory	Group theory
Hodge class t on A_h	$\mathbb{Q}$-subgroup $G_t \subseteq \mathrm{Sp}(V)$ fixing t
Hodge locus $\mathrm{NL}(t)$	union of orbits $H(\mathbb{R})^+ \cdot h$, $H \subseteq G_t$
very general point of a special subvariety	$\mathrm{Hg}(A_h) = H$
very general point of $\mathcal{A}_g$	$\mathrm{Hg}(A_h) = \mathrm{Sp}(V)$ (Theorem 5.2)

Example 4.10 ($\mathcal{A}_1$)**.** By Theorem 3.7, the possible Hodge groups are SL_2 and the tori U_K. Since $U_K(\mathbb{R}) \cong U(1)$ is compact, the orbit $U_K(\mathbb{R}) \cdot h$ is the single point $\{h\}$. So the special subvarieties of $\mathcal{A}_1$ are $\mathcal{A}_1$ itself and the CM points, and the Hodge loci (e.g. of the classes on $E \times E$ from Theorem 3.9) are the CM points. These are countable and dense. So in theorems about the Hodge classes on a "generic" member of a family, *general* (outside a proper Zariski closed subset) cannot be improved to anything better than *very general*.

Example 4.11 ($\mathcal{A}_2$)**.** By Lefschetz $(1,1)$, the Hodge loci of classes in $H^2(A, \mathbb{Q})$ are the loci where $\rho(A) \geq 2$; their components are called *Humbert surfaces*. There are two kinds of generic Hodge group.

(a) Split $V = V_1 \oplus V_2$ into ψ-orthogonal planes, so $\theta = \theta_1 + \theta_2$ with $\theta_i \in \Lambda^2 V_i$. Then $G_{\theta_1} = \mathrm{SL}(V_1) \times \mathrm{SL}(V_2)$, and $\mathrm{NL}(\theta_1) = \mathfrak{H}_1 \times \mathfrak{H}_1$, whose image in $\mathcal{A}_2$ is the locus of products $E_1 \times E_2$. Its generic Hodge group is $\mathrm{SL}_2 \times \mathrm{SL}_2 = (\mathrm{Sp}_{2,\mathbb{Q}})^2$ by Theorem 3.10.

(b) Let F be a real quadratic field acting on V, making V a 2-dimensional F-vector space with $\psi = \mathrm{Tr}_{F/\mathbb{Q}} \circ \psi_F$. Then $H = \mathrm{Res}_{F/\mathbb{Q}} \mathrm{SL}_2$ has $H(\mathbb{R}) = \mathrm{SL}_2(\mathbb{R}) \times \mathrm{SL}_2(\mathbb{R})$, the orbit is again $\mathfrak{H}_1 \times \mathfrak{H}_1$, and its image in $\mathcal{A}_2$ is a Hilbert modular surface, parametrizing abelian surfaces with real multiplication by F.

In both cases $H_{\mathbb{C}} \cong \mathrm{SL}_{2,\mathbb{C}} \times \mathrm{SL}_{2,\mathbb{C}}$ acting on $V_{\mathbb{C}}$ by the sum of the two standard representations, but only in case (a) is H a product over $\mathbb{Q}$.

5. The very general abelian variety

Lemma 5.1. *Let W be a $\mathbb{Q}$-vector space of dimension $2h$ and $\theta \in \Lambda^2 W$ nondegenerate. Then*

$$(\Lambda^\bullet W)^{\mathrm{Sp}(W)} = \mathbb{Q}[\theta]/(\theta^{h+1}).$$

Proof. By the Lefschetz decomposition (which is pure linear algebra here), $\Lambda^k W = \bigoplus_j \theta^j \wedge P^{k-2j}$, where $P^i \subseteq \Lambda^i W$ $(0 \leq i \leq h)$ is the primitive subspace. This decomposition is $\mathrm{Sp}(W)$-equivariant since θ is invariant. For $1 \leq i \leq h$, $P^i_{\mathbb{C}}$ is a nontrivial irreducible representation of $\mathrm{Sp}_{2h}(\mathbb{C})$ [FH91, Theorem 17.5], so it has no invariants. Only $P^0 = \mathbb{Q}$ contributes, and its copy in $\Lambda^{2j} W$ is spanned by θ^j. □

Theorem 5.2. *Let (A,θ) be a very general principally polarized abelian variety of dimension g. Then*

$$\mathrm{Hg}(A) = \mathrm{Sp}(V) \cong \mathrm{Sp}_{2g}, \qquad \mathrm{MT}(A) = \mathrm{GSp}(V), \qquad \mathrm{Hdg}^p(A) = \mathbb{Q}\cdot\theta^p.$$

In particular, A satisfies the Hodge conjecture.

Proof. For each of the countably many tensors $t \in T^{a,b}(V)$, $\mathrm{NL}(t)$ is a closed analytic subset of $\mathfrak{H}_g$. Take h outside every $\mathrm{NL}(t)$ that is not all of $\mathfrak{H}_g$. (Note that $\gamma\cdot\mathrm{NL}(t) = \mathrm{NL}(\gamma t)$ for $\gamma \in \mathrm{Sp}_{2g}(\mathbb{Z})$, so this is a condition on the image of h in $\mathcal{A}_g$.)

Claim: every Hodge tensor t for h is $\mathrm{Sp}(V)$-invariant. Indeed, $\mathrm{NL}(t) = \mathfrak{H}_g$, so t is fixed by $h'(U(1))$ for every $h' \in \mathfrak{H}_g$. Since $\mathfrak{H}_g = \{ghg^{-1} : g \in \mathrm{Sp}(V_{\mathbb{R}})\}$, this means $\mathrm{Stab}(t)$ contains every $\mathrm{Sp}(V_{\mathbb{R}})$-conjugate of $h(U(1))$. These generate a nontrivial connected normal subgroup of $\mathrm{Sp}(V_{\mathbb{R}})$, whose Zariski closure is all of $\mathrm{Sp}(V_{\mathbb{R}})$ because Sp_{2g} is simple. So $\mathrm{Sp}(V) \subseteq \mathrm{Stab}(t)$.

Now by Theorem 3.3, Theorem 3.5(a), and the claim,

$$(\Lambda^{2p}V)^{\mathrm{Sp}(V)} \subseteq (\Lambda^{2p}V)^{\mathrm{Hg}(A)} = \mathrm{Hdg}^p(A) \subseteq (\Lambda^{2p}V)^{\mathrm{Sp}(V)},$$

and the outer terms are $\mathbb{Q}\theta^p$ by Theorem 5.1. The class θ^p is algebraic: it is the class of $\Theta_1 \cap \cdots \cap \Theta_p$ for general translates Θ_i of a theta divisor. Finally, by Theorem 3.6, $\mathrm{Hg}(A)$ is the subgroup fixing all Hodge tensors, which by the claim contains $\mathrm{Sp}(V)$; combined with Theorem 3.5(a), $\mathrm{Hg}(A) = \mathrm{Sp}(V)$. □

Remark 5.3. By Theorem 4.10, and similarly because CM points are dense in $\mathcal{A}_g$, the words "very general" cannot be replaced by "general".

5.1. **Back to integral classes.**

Corollary 5.4. *For a very general principally polarized abelian variety (A,θ) of dimension g,*

$$\mathrm{Hdg}^p(A,\mathbb{Z}) = \mathbb{Z}\cdot\frac{\theta^p}{p!}.$$

Proof. In a symplectic basis, $\frac{\theta^p}{p!} = \sum_{|I|=p}\bigwedge_{i\in I}(e_i\wedge f_i)$, which lies in $\Lambda^{2p}H^1(A,\mathbb{Z})$ and is indivisible. So $\mathbb{Q}\theta^p \cap H^{2p}(A,\mathbb{Z}) = \mathbb{Z}\cdot\theta^p/p!$. □

So the integral Hodge conjecture for A in degree p is equivalent to $\theta^p/p!$ being algebraic (we already know $\theta^p = p!\cdot\theta^p/p!$ is). For $p=1$ this is clear. For curves, $p = g-1$, the class $\theta^{g-1}/(g-1)!$ is called the *minimal class*.

- If $(A,\theta) = (J(C),\theta)$ is a Jacobian, then the Abel–Jacobi image of C has class $\theta^{g-1}/(g-1)!$ (Poincaré's formula). Every principally polarized abelian variety of dimension $g \le 3$ is a product of Jacobians, so the minimal class is algebraic there.
- Conversely, by the Matsusaka–Ran criterion [Mat59, Ran81], if the minimal class is represented by an *effective* curve then (A,θ) is a product of Jacobians. Since $\dim\mathcal{M}_g = 3g-3 < g(g+1)/2 = \dim\mathcal{A}_g$ for $g \ge 4$, the minimal class on a very general A of dimension ≥ 4 is not effective. It could still be a difference of effective classes.
- For $g = 4, 5$ the very general principally polarized abelian variety is a Prym variety, and the Abel–Prym curve has class $2\theta^{g-1}/(g-1)!$.

Theorem 5.5 (Engel–de Gaay Fortman–Schreieder [EdGFS25])**.** *Let (A,θ) be a very general principally polarized abelian variety of dimension $g \ge 4$. Then the class of every curve on*

A is an even *multiple of $\theta^{g-1}/(g-1)!$. In particular, the minimal class is a non-algebraic integral Hodge class, and the integral Hodge conjecture fails for A.*

They prove the same for the intermediate Jacobian of a very general cubic threefold, which implies that very general cubic threefolds are not stably rational.

Remark 5.6 (Exceptional Hodge classes). The very general abelian variety has as few Hodge classes as possible. Special abelian varieties can have Hodge classes that are not polynomials in divisor classes, and for these the Hodge conjecture is a real problem. The classical example is due to Weil [Wei79]: if an imaginary quadratic field K acts on an abelian fourfold A so that each embedding of K occurs with multiplicity 2 on $H^{1,0}$, then $\Lambda^4_K H^1(A,\mathbb{Q}) \subseteq H^4(A,\mathbb{Q})$ is a 2-dimensional space of Hodge classes (*Weil classes*), which for very general such A are not generated by divisors. Proving that Weil classes are algebraic is a much-studied test case of the Hodge conjecture. See [Gor99] for a survey.

6. Lagrangian fibrations

Definition 6.1 ([BL22])**.** A *primitive symplectic variety* is a normal compact Kähler space X with a holomorphic symplectic form σ on X^{reg} that extends holomorphically to a resolution of X, such that $H^1(X,\mathcal{O}_X) = 0$ and $H^0(X,\Omega^{[2]}_X) = \mathbb{C}\sigma$. The smooth ones are exactly the irreducible hyperkähler manifolds.

Let X be a primitive symplectic variety of dimension $2n$. A *Lagrangian fibration* is a surjective morphism $f\colon X \to B$ with connected fibers onto a normal projective variety with $0 < \dim B < 2n$. Then $\dim B = n$, the fibers are Lagrangian, and the fiber X_b over a general point $b \in B$ is an abelian variety of dimension n (Matsushita [Mat99], in the smooth case).

Let $B^0 \subseteq B$ be the dense open subset over which f is smooth. Fixing an ample class on X and restricting it to the fibers polarizes every X_b, $b \in B^0$, with the same polarization type δ. So we get a moduli map

$$\phi\colon B^0 \to \mathcal{A}_{n,\delta}, \qquad b \mapsto [X_b],$$

where $\mathcal{A}_{n,\delta}$ is the moduli space of n-dimensional abelian varieties with a polarization of type δ (the analogue of $\mathcal{A}_n$ for non-principal polarizations).

Definition 6.2. The Lagrangian fibration f is *maximally varying* if ϕ is generically finite onto its image, i.e. the isomorphism class of X_b varies in all $n = \dim B$ directions.

Example 6.3. Let $S \to \mathbb{P}^1$ be an elliptic K3 surface whose j-invariant is not constant. Then

$$S^{[n]} \to \mathrm{Sym}^n S \to \mathrm{Sym}^n \mathbb{P}^1 \cong \mathbb{P}^n$$

is a Lagrangian fibration, whose fiber over $b = \{t_1,\dots,t_n\}$ (distinct points with smooth fibers E_{t_i}) is $E_{t_1} \times \cdots \times E_{t_n}$. Since j is not constant, this fibration is maximally varying. Since moreover isogeny classes are countable, for very general b the curves E_{t_i} have no CM and are pairwise non-isogenous. So by Theorem 3.10,

$$\mathrm{MT}(X_b) = \mathbb{G}_m \cdot (\mathrm{Sp}_{2,\mathbb{Q}})^n.$$

The main theorem says that this is what always happens, up to replacing Sp_2 by Sp_{2h}.

Theorem 6.4 (Varvak)**.** *Let X be a primitive symplectic variety with a maximally varying Lagrangian fibration $f\colon X \to B$. For very general $b \in B$,*

$$\mathrm{MT}(X_b) \cong \mathbb{G}_m \cdot (\mathrm{Sp}_{2h,\mathbb{Q}})^k$$

for some $h, k \geq 1$. Equivalently, $\mathrm{Hg}(X_b) \cong (\mathrm{Sp}_{2h,\mathbb{Q}})^k$.

(For the equivalence: if $\mathrm{MT} = \mathbb{G}_m \cdot S$ with S semisimple, then $S = [\mathrm{MT}, \mathrm{MT}] = [\mathrm{Hg}, \mathrm{Hg}] \subseteq \mathrm{Hg}$, and Hg is a connected subgroup of $(\mathrm{MT} \cap \mathrm{SL}(V))$, whose identity component is S.)

Remark 6.5 (The maximal variation hypothesis is necessary). Let $A = E \times F$ be a product of elliptic curves and $X = K_n(A) \subseteq A^{[n+1]}$ the generalized Kummer variety. The projection $A \to E$ induces a Lagrangian fibration $X \to \mathbb{P}^n$, and over a dense open subset of $\mathbb{P}^n$ every fiber is isomorphic to

$$\left\{(y_1, \ldots, y_{n+1}) \in F^{n+1} : \sum_i y_i = 0\right\} \cong F^n.$$

So this fibration is isotrivial (ϕ is constant). If F has CM by K, then $\mathrm{MT}(X_b) = \mathrm{MT}(F) = \mathrm{Res}_{K/\mathbb{Q}} \mathbb{G}_m$ is a torus, not of the form in Theorem 6.4. (The Hodge conjecture still holds for F^n: the Hodge ring of a power of an elliptic curve is generated by divisor classes.) Matsushita conjectured that for a Lagrangian fibration of a hyperkähler manifold, ϕ is either constant or generically finite onto its image, so conjecturally the isotrivial fibrations are the only ones excluded.

Corollary 6.6. *In the situation of Theorem 6.4, for very general $b \in B$ the abelian variety X_b, and every power X_b^m, satisfies the Hodge conjecture.*

The proof of Theorem 6.4 is not the topic of this talk. Instead, let us see how the corollary follows. It is a statement purely about abelian varieties:

Theorem 6.7. *Let A be an abelian variety with $\mathrm{Hg}(A) \cong (\mathrm{Sp}_{2h,\mathbb{Q}})^k$. Then every Hodge class on A is a polynomial in divisor classes. In particular, A satisfies the Hodge conjecture, and so does every power A^m.*

The proof has two steps. First we pin down how $\mathrm{Hg}(A)$ acts on $V = H^1(A, \mathbb{Q})$, and then we compute invariants.

Lemma 6.8. *Write $\mathrm{Hg}(A) = S_1 \times \cdots \times S_k$ with $S_i \cong \mathrm{Sp}_{2h,\mathbb{Q}}$, and let W_i be the standard $2h$-dimensional representation of S_i. Let $V^{(i)} = V^{\prod_{j \neq i} S_j}$. Then*

$$V = \bigoplus_{i=1}^{k} V^{(i)}, \qquad V^{(i)} \cong W_i \otimes M_i,$$

where M_i is a vector space with trivial action, of dimension m_i.

Proof. Let $\mu = h|_{U(1)} \colon U(1) \to \mathrm{Hg}(A)(\mathbb{R})$, with components $\mu_i \colon U(1) \to S_i(\mathbb{R})$.

(a) *Each μ_i is nontrivial.* Otherwise $h(U(1)) \subseteq \prod_{j \neq i} S_j$, a smaller $\mathbb{Q}$-subgroup, contradicting the minimality of $\mathrm{Hg}(A)$. (This is where we use that the factors S_i are defined over $\mathbb{Q}$.)

(b) *Each irreducible constituent of $V_{\mathbb{C}}$ is a representation of a single factor.* Let $U = U_1 \boxtimes \cdots \boxtimes U_k$ be an irreducible constituent. The torus $\mu(U(1))$ acts on $V_{\mathbb{C}}$ with weights exactly ± 1 (by z on $V^{1,0}$ and $\overline{z} = z^{-1}$ on $V^{0,1}$). If U_i is nontrivial, then μ_i acts on U_i nontrivially (as $S_i \to \mathrm{GL}(U_i)$ has finite kernel) with determinant 1, so with weights $a < 0 < b$ among others. If U_i and U_j were both nontrivial for some $i \neq j$, with μ_j-weights $c < 0 < d$, then μ would have the three distinct weights $a + c < a + d < b + d$ on U. This is impossible, and U cannot be trivial (weight 0), so $U = U_i$ for exactly one i, and μ_i acts on U_i with weights exactly ± 1.

(c) *U_i is the standard representation.* Since μ_i has weights ± 1 on U_i, it has weights in $\{0, \pm 2\}$ on $\mathfrak{sp}_{2h} \subseteq \mathrm{End}(U_i)$. In standard coordinates, with μ_i dominant, this forces $\mu_i = (1, \dots, 1)$. If λ is the highest weight of U_i, then the weights λ and $(-\lambda_1, \lambda_2, \dots, \lambda_h)$ pair with μ_i to $\sum_j \lambda_j \in \{\pm 1\}$ and $\sum_j \lambda_j - 2\lambda_1 \in \{\pm 1\}$, which forces $\lambda = (1, 0, \dots, 0)$.

(d) So $V_{\mathbb{C}}^{(i)}$ is isotypic of type $W_{i,\mathbb{C}}$ and $V = \bigoplus_i V^{(i)}$. Since W_i is defined over $\mathbb{Q}$ and absolutely irreducible, $V^{(i)} \cong W_i \otimes \mathrm{Hom}_{S_i}(W_i, V^{(i)})$. □

Remark 6.9. By Theorem 3.5, the $V^{(i)}$ are sub-Hodge structures and $\mathrm{End}_{\mathrm{HS}}(V^{(i)}) = \mathrm{End}(M_i)$. So $A \sim A_1^{m_1} \times \cdots \times A_k^{m_k}$ up to isogeny, with the A_i pairwise non-isogenous abelian varieties of dimension h satisfying $\mathrm{Hg}(A_i) = \mathrm{Sp}_{2h}$, and $\dim A = h \sum_i m_i$. In Theorem 6.3, $h = 1$, $k = n$, and all $m_i = 1$.

Remark 6.10 (Why splitting over $\mathbb{Q}$ matters). The abstract group $\mathrm{Hg}(A)_{\mathbb{C}}$ does not determine the Hodge classes. Mumford [Mum69] constructed abelian fourfolds with $\mathrm{End}(A) = \mathbb{Z}$ and $\mathrm{Hg}(A)_{\mathbb{C}} \cong \mathrm{SL}_{2,\mathbb{C}}^3$, acting on $V_{\mathbb{C}} \cong \mathbb{C}^2 \otimes \mathbb{C}^2 \otimes \mathbb{C}^2$. There $\mathrm{Hg}(A)$ is a non-split $\mathbb{Q}$-form of SL_2^3, two of the three factors of $\mathrm{Hg}(A)_{\mathbb{R}}$ are compact, μ is trivial on them, and step (a) above fails. Similarly, Theorem 4.11(b) shows that $\mathrm{Hg}(A)_{\mathbb{C}} \cong \mathrm{Sp}_{2h,\mathbb{C}}^k$ does not imply $\mathrm{Hg}(A) \cong \mathrm{Sp}_{2h,\mathbb{Q}}^k$.

The second step is classical invariant theory.

Lemma 6.11 (First fundamental theorem for Sp, skew version [How89])**.** *Let W be a symplectic $\mathbb{Q}$-vector space with symplectic class θ, and M a finite-dimensional $\mathbb{Q}$-vector space with trivial $\mathrm{Sp}(W)$-action. Then the algebra $(\Lambda^\bullet(W \otimes M))^{\mathrm{Sp}(W)}$ is generated by its degree 2 part, which is $\theta \otimes \mathrm{Sym}^2 M \subseteq (\Lambda^2 W \otimes \mathrm{Sym}^2 M) \oplus (\mathrm{Sym}^2 W \otimes \Lambda^2 M) = \Lambda^2(W \otimes M)$.*

When $\dim M = 1$ this is just Theorem 5.1.

Proof of Theorem 6.7. By Theorem 6.8,

$$H^\bullet(A, \mathbb{Q}) = \Lambda^\bullet V = \bigotimes_{i=1}^k \Lambda^\bullet(W_i \otimes M_i),$$

and S_i acts only on the i-th factor. Hence by Theorem 3.3,

$$\mathrm{Hdg}^\bullet(A) = \left(\Lambda^\bullet V\right)^{S_1 \times \cdots \times S_k} = \bigotimes_{i=1}^k \left(\Lambda^\bullet(W_i \otimes M_i)\right)^{S_i}.$$

By Theorem 6.11, this algebra is generated by its degree 2 part, which consists of Hodge classes in $H^2(A, \mathbb{Q})$. By Lefschetz $(1,1)$ these are classes of $\mathbb{Q}$-divisors, and products of divisor classes are algebraic (intersect general translates). So every Hodge class on A is algebraic.

For A^m: by Theorem 3.2(4), $\mathrm{Hg}(A^m) = \mathrm{Hg}(A)$, now acting on $V^{\oplus m} = \bigoplus_i W_i \otimes (M_i \otimes \mathbb{Q}^m)$, and the same argument applies. □

Remark 6.12. When all $m_i = 1$ (as in Theorem 6.3), only Theorem 5.1 is needed: the Hodge ring of A is $\bigotimes_i \mathbb{Q}[\theta_i]/(\theta_i^{h+1})$, where $\theta_i \in \Lambda^2 W_i \subseteq H^2(A, \mathbb{Q})$ is the pullback of a polarization along $A \to A_i$. This is exactly parallel to Theorem 5.2, with Sp_{2h}^k in place of Sp_{2g}.

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